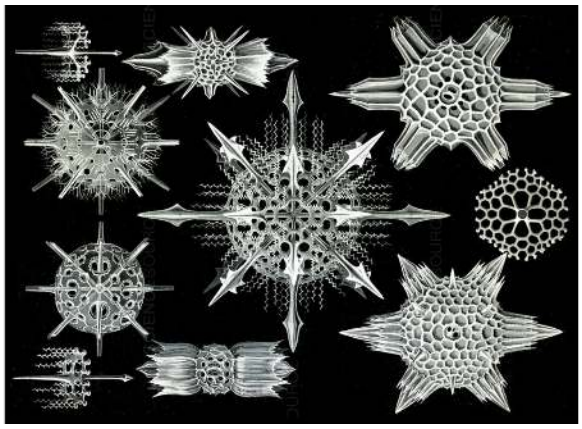
EART11100 — Evolution 101


Russell Garwood | russell.garwood@manchester.ac.uk
Office: Williamson 1.56

• An introduction to molecular (DNA) and morphological (anatomy) evolution •

• Variation – sources and its heritable nature •

• Competition, and fitness in the context of evolution •

• Natural selection •

• Speciation, and evolutionary (phylogenetic) trees •

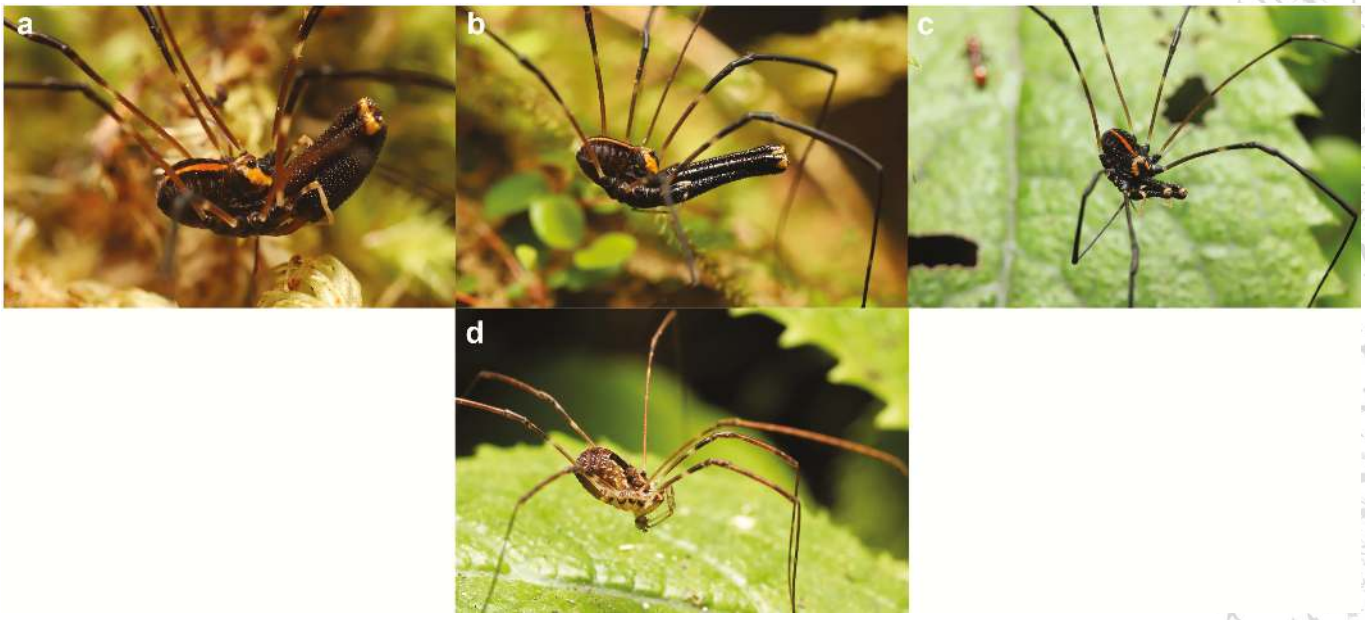
Why does it matter?





Nothing in Biology Makes Sense Except in the Light of Evolution

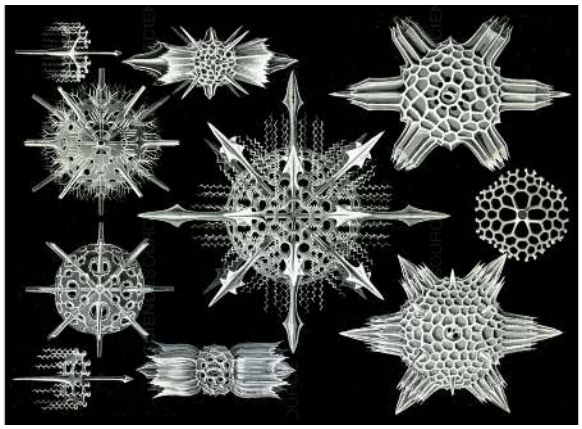
Theodosius Dobzhansky



Source: Powell, E.C., Painting, C.J., Hickey, A.J. and Holwell, G.I., 2020. Defining an intrasexual male weapon polymorphism in a New Zealand harvestman (Opiliones: Neopilionidae) using traditional and geometric morphometrics. Biological Journal of the Linnean Society, 130(2): 395-409.

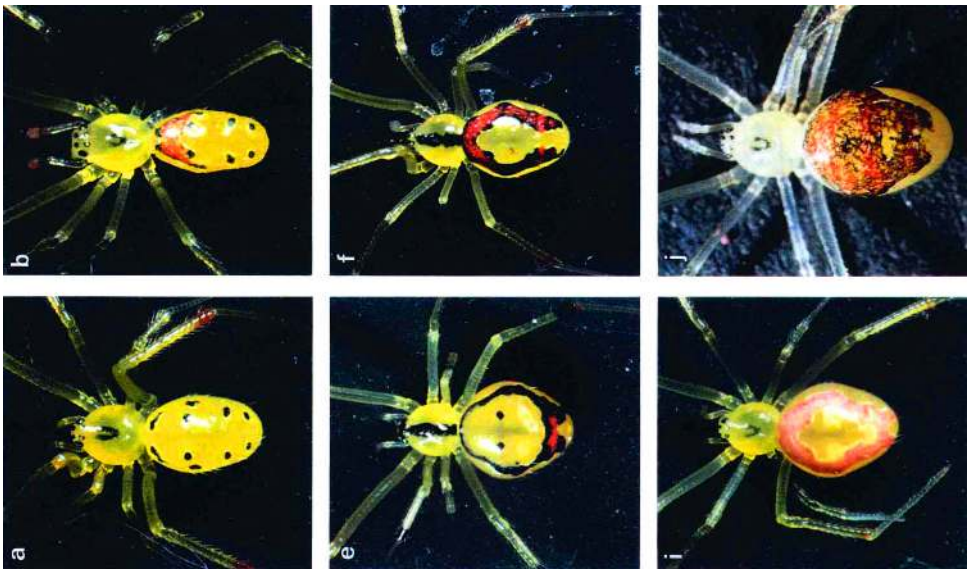
An introduction to molecular (DNA) and morphological (anatomy) evolution





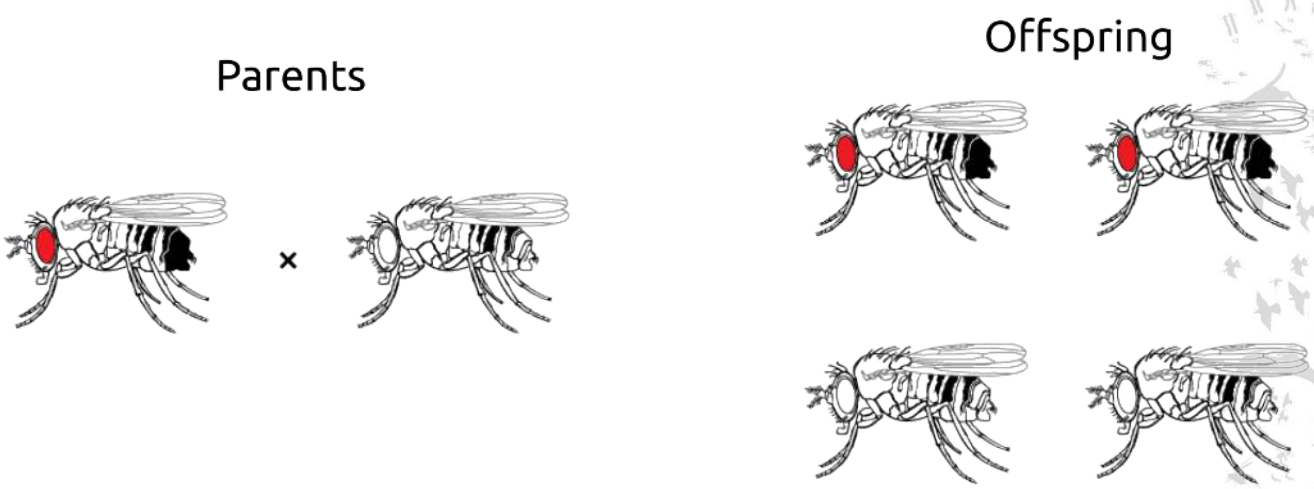


What do you need for evolution to occur?



Variation

Colour morphs of *Theridion grallator*, the Hawaiian happy-face spider.
Source: Oxford, G.S. and Gillespie, R.G., 1996. Genetics of a colour polymorphism in *Theridion grallator* (Araneae: Theridiidae), the Hawaiian happy-face spider, from Greater Maui. Heredity, 76(3), p.238.



Heredity

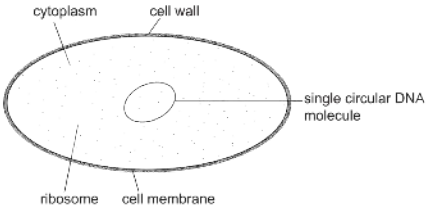
Source: Image modified after openStax



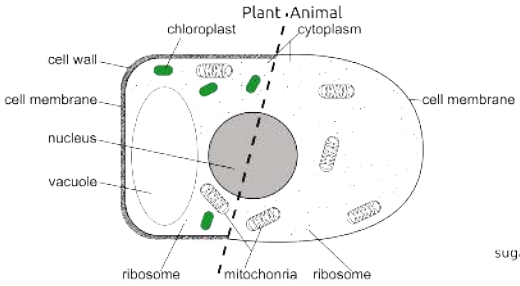
Selection

Sources: Alex Wild (left); Katja Schulz (right); CC0/CCBY.

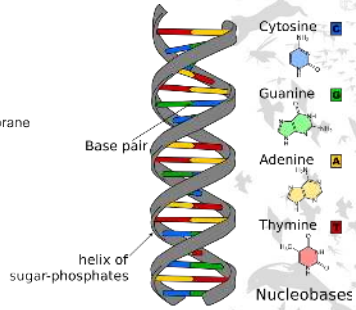
On what does evolution act?



Prokaryotes



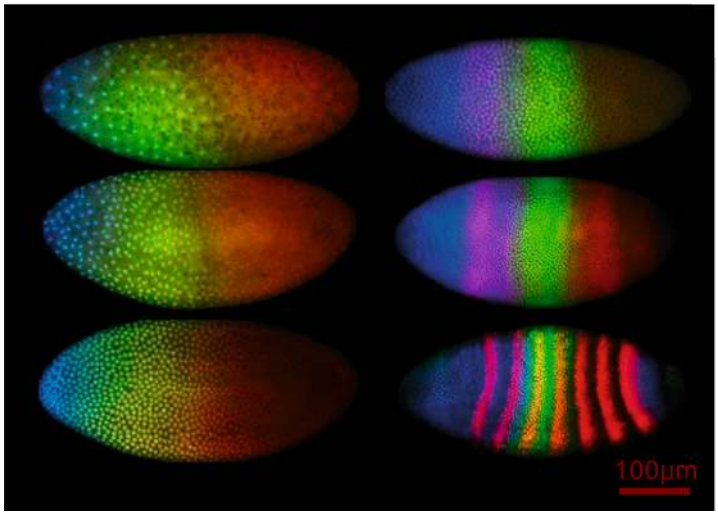
Eukaryotes

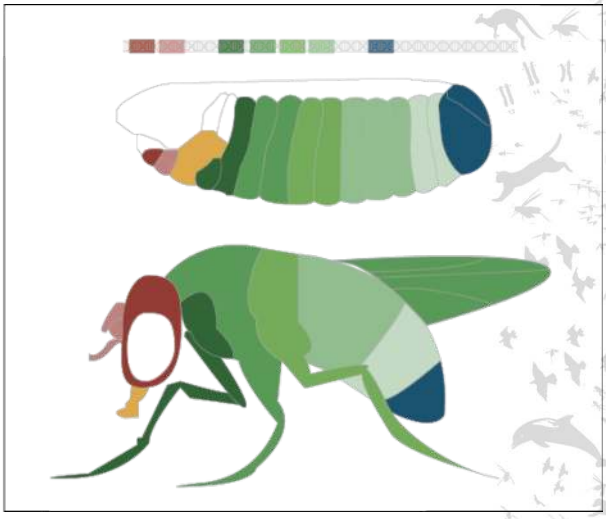


DNA

Genomes

Genome: The total genetic information an individual inherits from its parents.
Morphology: The form and structure of individual organisms.
Source for all definitions: Allaby, M., 2014. A dictionary of zoology. OUP Oxford.



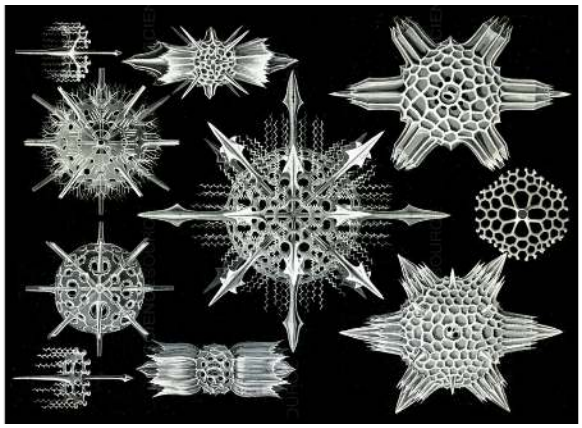


Genotype and Phenotype

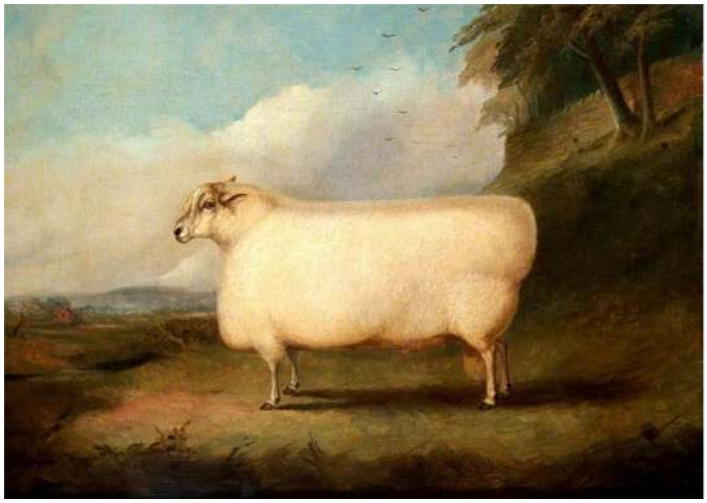
Genotype: The genetic constitution of an organism, as opposed to its physical appearance (phenotype)
Phenotype: The observable manifestation of a specific genotype; those properties of an organism produced by the genotype in conjunction with the environment that are observable.
Ontogeny: The development of an individual from fertilization of the egg to adulthood.

Sources: Fritsch, C., Ploeger, G. and Arndt-Jovin, D.J., 2006. Drosophila under the lens: imaging from chromosomes to whole embryos. Chromosome Research, 14(4), 451-464. / Stanford.edu

Variation – sources and its heritable nature

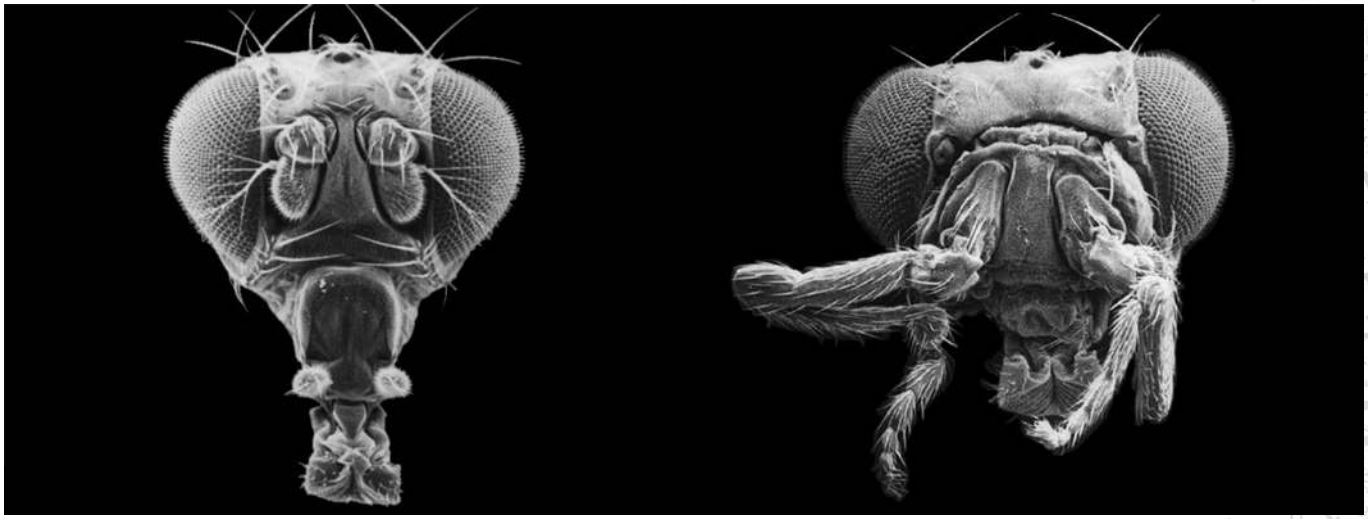




When does variation matter?




Heritability

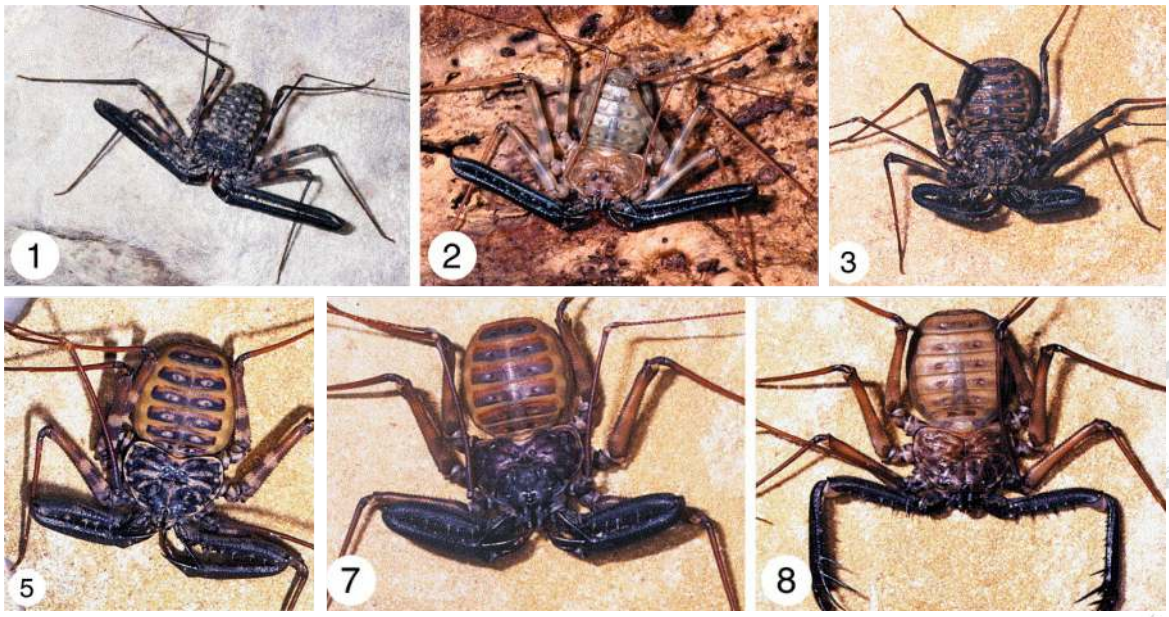
Whence variation?



Mutation

Images copyright Dr FR Turner

What have we covered so far?

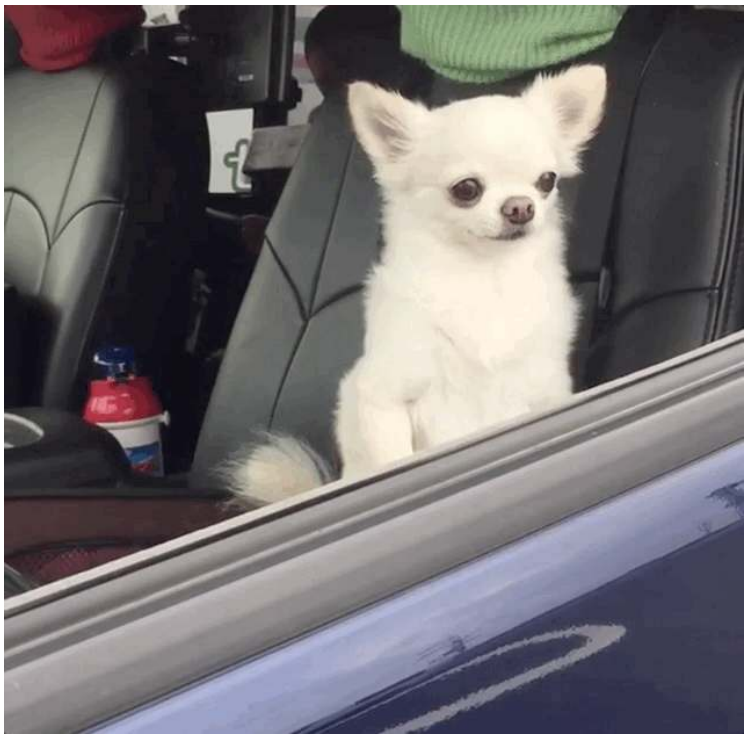


Source: Prendini, L., Weygoldt, P. and Wheeler, W.C., 2005. Systematics of the *Damon variegatus* group of African whip spiders (Chelicerata: Amblypygi): evidence from behaviour, morphology and DNA. Organisms Diversity & Evolution, 5(3), 203-236.

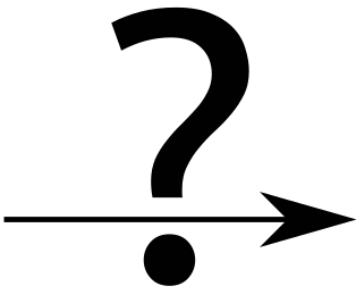
A quick thought experiment.



Image copyright Martin Cathrae CCBY









Discuss in small groups how you could turn a pack of wolves - if you had infinite time - into Chihuahuas?
If you have any questions about the material so far, please do ask Russell or the GTAs!

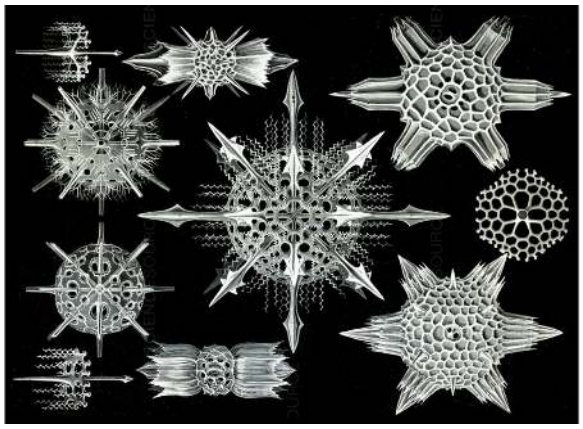
Thought experiment - Discussion



12/10, nice collar.

Competition, and fitness in the context of evolution







What is fitness?



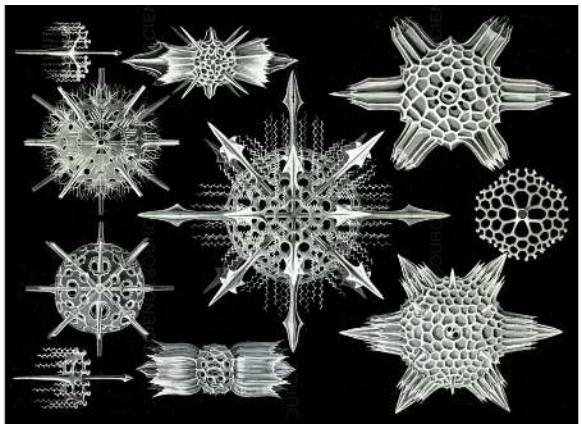


Fitness: The fitness of an individual is a measure of its ability, relative to others, to leave viable offspring.
Fecundity: The production of offspring.
Left: Indochinese Forest Lizard (CCBY, Bigtreebirds); Right: Ornate tree lizard (Steven Kessel).



Source: Baym, M., Lieberman, T.D., Kelsic, E.D., Chait, R., Gross, R., Yelin, I. and Kishony, R., 2016. Spatiotemporal microbial evolution on antibiotic landscapes. Science, 353(6304), 1147-1151.

Natural selection


What is natural selection?




Selection: The process that determines the relative share of different genotypes which individuals possess and propagate in a population. [...] Selection may be natural (e.g. by nature) or artificial (i.e. by human action, e.g. plant or animal breeders).

Natural selection: A complex process in which the total environment determines which members of a species survive to reproduce and so pass on their genes to the next generation. This need not necessarily involve a struggle between organisms.

Source: Ian Alexander CCBY.



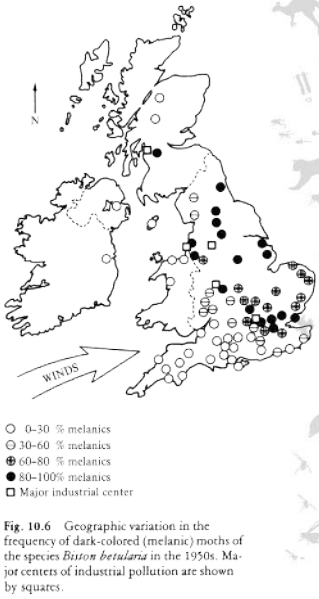


Fig. 10.6 Geographic variation in the frequency of dark-colored (melanic) moths of the species *Biston betularia* in the 1950s. Major centers of industrial pollution are shown by squares.

More: van't Hof, A.E., Campagne, P., Rigden, D.J., Yung, C.J., Lingley, J., Quail, M.A., Hall, N., Darby, A.C. and Saccheri, I.J., 2016. The industrial melanism mutation in British peppered moths is a transposable element. Nature, 534(7605), p.102.

Source: Ford, E.B., 1977. Ecological genetics. Springer, Dordrecht.

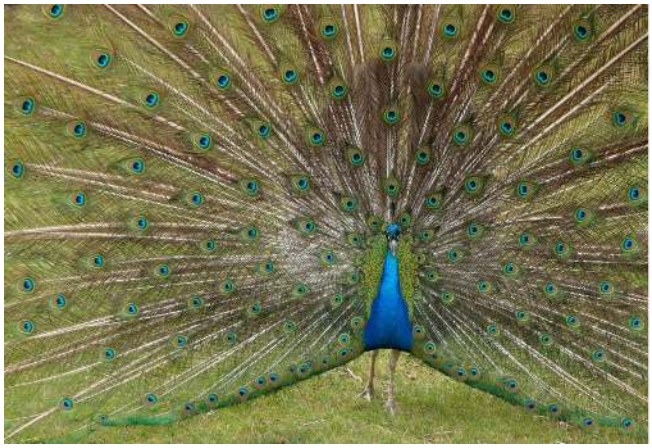
What is the response to natural selection?



Adaptation: Generally, the adjustments that occur in animals in respect of their environments. The adjustments may occur by natural selection, as individuals with favourable genetic traits breed more prolifically than those lacking these traits (genotypic adaptation), or they may involve non-genetic changes in individuals, such as physiological modification (e.g. acclimatization) or behavioural changes (phenotypic adaptation).

Giant Leaf Insect (Phyllium giganteum). Source: Bernard Dupont CCBY.

What other forms of selection are there?



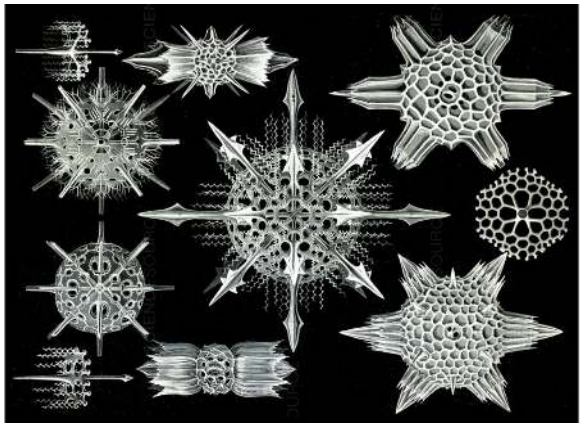


Sexual selection

Source: Paul Lakin (left) CCBY, CC0 (right).

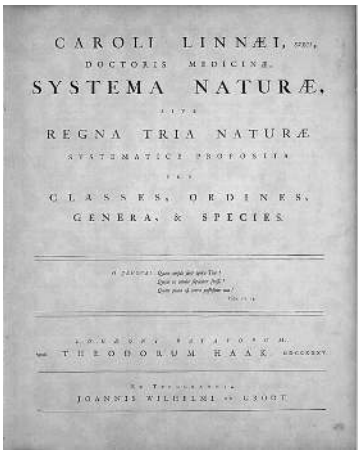
Speciation, and evolutionary (phylogenetic) trees

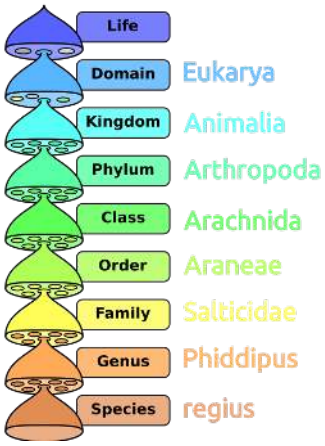






What are species?


CAROLI LINNÆI,
DOCTORIS MEDICINÆ,
SYSTEMA NATURÆ,
REGNA TRIA NATURÆ
CLASSES, ORDINES,
GENERA, & SPECIES.
THEODORUM HAAK.
JOANNIS WILHELMI DE GROOT.



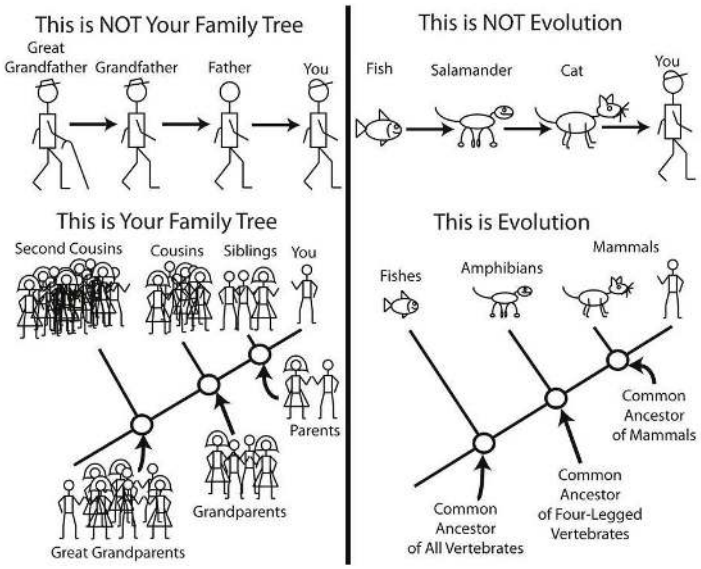


Source: Thomas Shahan CCBY.



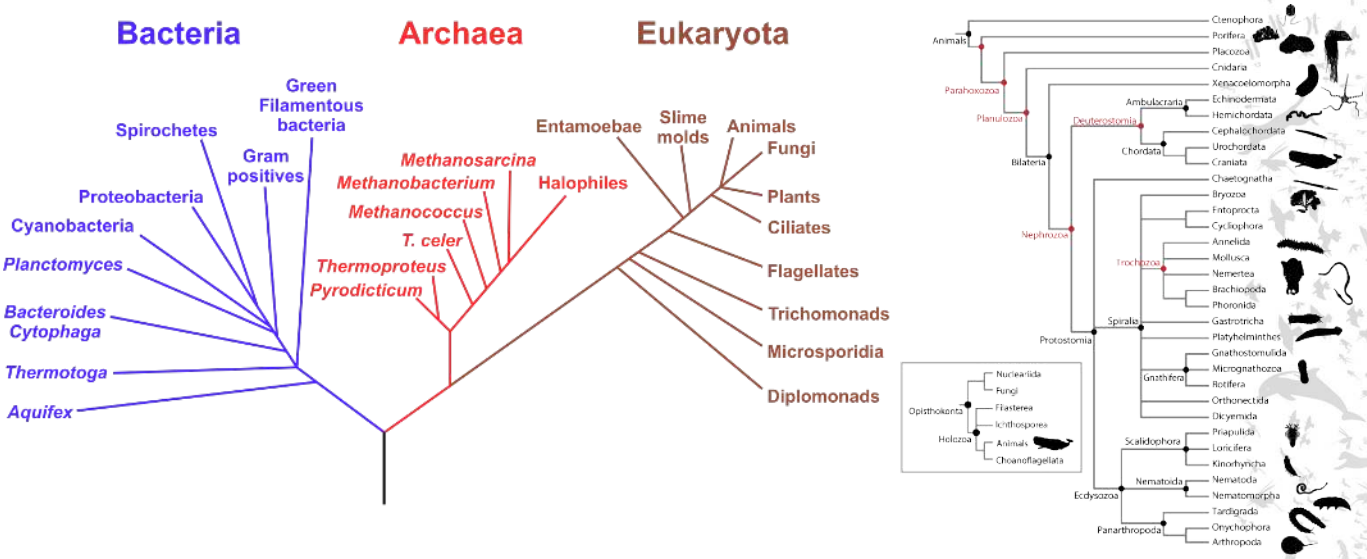


What is a phylogeny



Phylogeny: Evolutionary relationships within and between taxonomic levels, particularly the patterns of lines of descent, often branching, from one organism to another (i.e. the relationships of groups of organisms as reflected by their evolutionary history).

Source: Orfinger, A.B., 2015. Evolution as opinion versus fact: a false dichotomy with true consequences. EvoS J, 7(1), 13-20.



Source: Dunn, C.W., Giribet, G., Edgecombe, G.D. and Hejnol, A., 2014. Animal phylogeny and its evolutionary implications. Annual review of ecology, evolution, and systematics, 45, 371-395.

Exercise





In small groups, discuss the question on the sheet you have been handed regarding an adaptation of an organism.
What is the potential selective pressure for this feature?
Our wonderful GTAs are here to help you.

Scan the QR code for more!

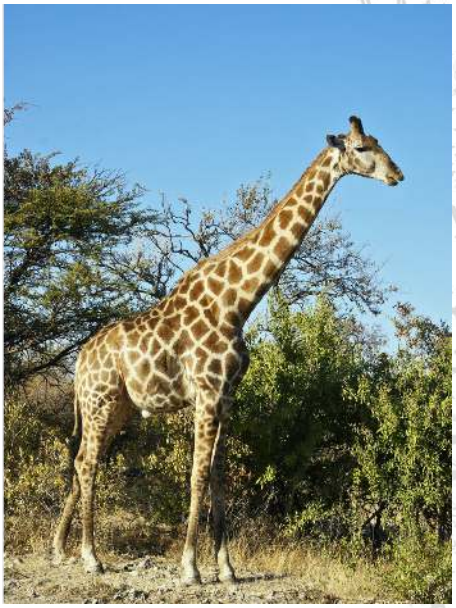




As with many things in life, and most in science, there isn't necessarily a correct answer.

Why the long neck?





Giraffe. Sources: Anand Desai and Hans Hillewaert CCBY

Simmons, R.E. and Altwegg, R., 2010. Necks-for-sex or competing browsers? A critique of ideas on the evolution of giraffe. Journal of zoology, 282(1), 6-12.

Why do snakes not have limbs?



Source: David Vraju CCBY
Wiens, J.J. and Slingluff, J.L., 2001. How lizards turn into snakes: a phylogenetic analysis of body-form evolution in anguid lizards. Evolution, 55(11), 2303-2318.

Why does a zebra have stripes?



Source: Sajjad Fazel CCBY
Caro, T., Izzo, A., Reiner Jr, R.C., Walker, H. and Stankowich, T., 2014. The function of zebra stripes. Nature communications, 5, p.3535.

Why the long finger?



Aye-aye. Source: Nomis-simon CCBY
Sterling, E.J., 1994. Aye-ayes: specialists on structurally defended resources. Folia Primatologica, 62(1-3), 142-154.



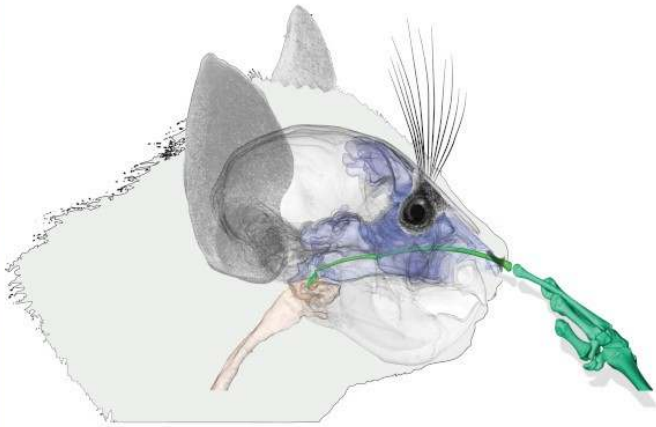


Image © Anne-Claire Fabre/Renaud Boistel

Why are chillis hot?



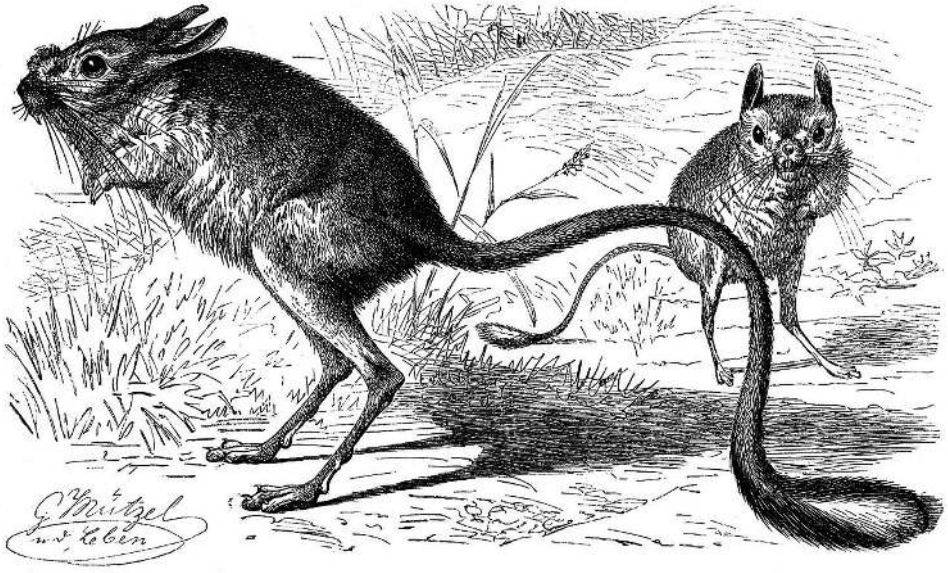
Chilli pepper.
Tewksbury, J.J. and Nabhan, G.P., 2001. Seed dispersal: directed deterrence by capsaicin in chillies. Nature, 412(6845), p.403.

Why are polar bears white?



Source: Arturo de Frias Marques CCBY
Harington, C.R., 2008. The evolution of Arctic marine mammals. Ecological Applications, 18(sp2), S23-S40.

Why do many mice have big ears?



This is a jerboa.

Knight, M.H. and Skinner, J.D., 1981. Thermoregulatory, reproductive and behavioural adaptations of the big eared desert mouse, Malacothrix typica to its arid environment. Journal of Arid Environments, 4(2), 137-145.

Why do elephants have big ears?



Elephant

Williams, T.M., 1990. Heat transfer in elephants: thermal partitioning based on skin temperature profiles. Journal of Zoology, 222(2), 235-245

Why do deer have antlers?



Source: Alastair Rae CCBY
Clutton-Brock, T.H., 1982. The functions of antlers. Behaviour, 79(2-4), 108-124.

Why do humans have big brains?





Human (this particular human is Grace Hopper, who had a *formidable* brain).
González-Forero, M. and Gardner, A., 2018. Inference of ecological and social drivers of human brain-size evolution. Nature, 557(7706), p.554.

Want to learn more?

Earth and Planetary Science: Palaeobiology

Environmental Science: Ecology, evolution and conservation biology

